

[This question paper contains 4 printed pages]

Your Roll No. :

Sl. No. of Q. Paper : 6636 **HC**

Unique Paper Code : 32351602

Name of the Course : **B.Sc.(Hons.)Mathematics**

Name of the Paper : Ring Theory and Linear
Algebra-II

Semester : VI

Time : 3 Hours **Maximum Marks : 75**

Instructions for Candidates :

- (a) Write your Roll No. on the top immediately on receipt of this question paper.
- (b) Attempt any **two** parts from each of the question.

1. (a) State and prove Division Algorithm for $F[x]$ where F is a field.

(b) Let p be a prime and $f(x) \in \mathbf{Z}_p[x]$ be irreducible over $\mathbf{Z}_p$ such that $\deg f(x) = n$,

prove that $\frac{\mathbf{Z}_p[x]}{\langle f(x) \rangle}$ is a field with p^n elements.

P.T.O.

- (c) In $\mathbb{Z}[\sqrt{-5}]$, prove that $1 + 3\sqrt{-5}$ is irreducible but not prime. 6, 6, 6
2. (a) Prove that a polynomial of degree n over a field F has at most n zeros, counting multiplicity. Give an example to show that the result does not hold if F is not a field.
- (b) Let F be a field and $p(x) \in F[x]$. Then prove that $\langle p(x) \rangle$ is a maximal ideal in $F[x]$ if and only if $p(x)$ is irreducible over F .
- (c) Prove that the ring of Gaussian integers $\mathbb{Z}(i)$ is a Euclidean domain. 6.5, 6.5, 6.5
3. (a) Let $V = P_1(1+R)$ and for $p(x) \in V$, define
- $$f_1, f_2 \in V^* \quad \text{by} \quad f_1(p(x)) = \int_0^1 p(t) dt \quad \text{and}$$
- $$f_2(p(x)) = \int_0^2 p(t) dt$$
- Prove that f_1, f_2 is a basis for V^* and find a basis for V for which it is the dual basis. 6
- (b) Prove that a linear operator T on a finite dimensional vector space V is diagonalizable if and only if there exists an ordered basis β for V consisting of eigenvectors of T .

- (c) Let $A = \begin{pmatrix} 0 & -2 \\ 1 & 3 \end{pmatrix}$ Show that A is diagonalizable and find a 2×2 matrix Q such that $Q^{-1}AQ$ is a diagonal matrix.

6, 6, 6

4. (a) Let T be a linear operator on a finite dimensional vector space V and W be a T invariant subspace of V . Prove that the characteristic polynomial of T_W divides the characteristic polynomial of T .
- (b) Let T be a linear operator on an n dimensional vector space V such that V is a T cyclic subspace of itself. Prove that $f(t)$, the characteristic polynomial of T and $p(t)$ the minimal polynomial of T have the same degree.
- (c) Let T be a linear operator $P_2(\mathbb{R})$ on such that $T(g(x)) = g'(x) + 2g(x)$. Find minimal polynomial of T . Hence, also determine if T is diagonalizable.

6.5, 6.5, 6.5

5. (a) State and prove Cauchy Schwarz inequality in an inner product space V over F and use it to prove triangle inequality in $\bar{V}$.

- (b) Apply the Gram-Schmidt process to the given subset S of the inner product space V to obtain an orthonormal basis β for $\text{span}(S)$ and compute the Fourier coefficients of the given vector x relative to β .

$$V = \mathbb{R}^3, S = \{(1,1,1), (0,1,1), (0,0,1)\} \text{ and } x = (1,0,1)$$

- (c) Let V be a finite dimensional inner product space and let T be a linear operator on V . Show that there exists a unique linear operator $T^*: V \rightarrow V$ such that

$$\langle T(x), Y \rangle = \langle x, T^* y \rangle \text{ for all } x, y \in V.$$

6.5, 6.5, 6.5

6. (a) For the set of data $\{(-3,9), (-2,6), (0,2), (1,1)\}$, use the least squares approximation to find the best fit with a linear function. Also compute the error E .
- (b) State and prove Schur Theorem.
- (c) Let V be an inner product space and let T be a linear operator on V . Then prove that T is an orthogonal projection if and only if T has an adjoint T^* and $T^2 = T = T^*$.

6, 6, 6